

MULTI-DIMENSIONAL PROSODY JUDGMENT FOR LIVE STREAMING SPEECH SYNTHESIS

Zifan Guan^{1,2}, Longyu Lu², Junan Zhang¹, Zhizheng Wu¹, Meiguang Jin², Junfeng Ma²

¹The Chinese University of Hong Kong, Shenzhen, China

²TaoLive-AIGC Team

Taobao & Tmall Group of Alibaba

ABSTRACT

Evaluating live streaming speech synthesis (TTS) requires assessing fine-grained, highly expressive prosody—such as emotion, intonation, and energy—which traditional MOS predictors fail to capture. While proprietary Large Language Models (LLMs) like Gemini can evaluate these aspects, they are too costly for massive inference and reinforcement learning feedback. To address this, we first introduce Live-ProsodyJudge (LPJ), a cost-effective pairwise evaluator distilled from Gemini into Qwen3-Omni.

However, we identify a critical flaw in standard multi-dimensional evaluation: verdict coupling. The judge tends to lazily align all individual dimension scores with its overall preference, collapsing a rich multi-dimensional rubric into a single preference bit. To resolve this, we further propose Decoupled-Live-ProsodyJudge (D-LPJ). D-LPJ eliminates the overall verdict target to prevent blind following, masks uncertain pair-dimensions during Supervised Fine-Tuning (SFT), and introduces a novel span-local GRPO strategy that applies normalized advantages strictly to their corresponding rationale spans.

Evaluated on highly curated human-annotated test sets, 10-sample balanced-order LPJ achieves higher point accuracy than a single Gemini call, while D-LPJ successfully produces independent, decoupled dimension judgments. Furthermore, in a Best-of-8 TTS candidate selection tournament, the LPJ-selected utterance falls within the human top-3 in 85.29% of high-confidence cases, demonstrating its efficacy for fine-grained TTS preference optimization.

Index Terms— prosody evaluation, expressive speech evaluation, speech reward model, reinforcement learning

1. INTRODUCTION

Live streaming TTS requires exceptional expressiveness: fluency, varied intonation, appropriate emotion, high energy, and natural shifts between product explanation and audience interaction. Traditional reference-free Mean Opinion Score (MOS) predictors [1, 2, 3] regress to naturalness or transmission quality, inherently failing to capture these complex interaction-mode and expressive phenomena.

Recent expressive and controllable TTS systems such as dots.tts [4] make fine-grained evaluation increasingly important. Among existing benchmarks, EmergentTTS-Eval [5] examines complex prosodic, expressive, and linguistic challenges with model-as-a-judge, while long-form speech benchmarking [6] covers diverse generation scenarios and CEAval [7] tests expressive appropriateness in rich contexts. SpeechJudge [8] assesses speech naturalness, whereas GSRM [9] and UniSRM [10] provide generative or

reasoning-based fine-grained rewards for speech RLHF. However, these methods do not target live-streaming delivery and interaction.

Powerful audio-language models such as Gemini-3.1-pro-preview [11] can make difficult expressive judgments, but proprietary inference is prohibitively expensive for iterative TTS evaluation and reinforcement learning optimization (RLHF) over thousands of sampled utterances. We therefore distill Gemini’s reasoning into open-weight Qwen3-Omni [12], establishing the baseline Live-ProsodyJudge (LPJ) under a seven-dimension protocol: four always-scored core dimensions (C1–C4)—fluency, intonation, emotion, and live streaming expressiveness—and three transcript-triggered dimensions (L1–L3)—key-information emphasis, emotional-state switching, and interaction-mode switching.

Despite achieving reasonable accuracy through swap-consistent distillation and curriculum learning, we observe a severe systematic failure in the baseline LPJ: verdict coupling. In a nominally multi-dimensional evaluation, the model often collapses its judgments to follow the overall winner. For instance, if utterance A is deemed better overall, the model lazily assigns wins to utterance A across fluency, intonation, emotion, and expressiveness, severely limiting its utility for targeted TTS optimization where distinct prosodic dimensions must yield distinct verdicts.

To force dimension-independent evaluation, we introduce Decoupled-Live-ProsodyJudge (D-LPJ). Because collapse is most severe in the four dimensions scored for every pair, while the conditional L1–L3 dimensions provide sparse supervision, D-LPJ focuses on C1–C4. It removes the overall-verdict target and masks uncertain pair-dimensions during SFT rather than discarding the entire pair. Finally, span-local GRPO [13] applies each dimension’s advantage only to its own rationale span, preventing cross-dimension reward cancellation.

This work makes three main contributions:

Protocol & Data Formulation: We define a multi-dimensional prosody rubric for live streaming TTS and construct an evaluation suite of 1,043 internal human-annotated pairs (plus a 222-pair transfer set), strictly controlling text transcripts to eliminate lexical confounding.

Dimension-Decoupling Architecture: We propose D-LPJ, which mitigates verdict coupling by eliminating the overall-verdict target, employing dimension-wise SFT masking, and introducing span-local GRPO for fine-grained credit assignment.

Empirical Validation: Our results demonstrate that D-LPJ avoids verdict collapse and yields varying multi-dimensional verdicts. Under 10-sample balanced-order inference, LPJ exhibits negligible position bias and achieves higher point agreement than a single Gemini call on four main test sets, while D-LPJ achieves 86.10% pooled agreement. In practical Best-of-8 candidate selection, LPJ’s

top choice matches human top-3 preferences in 85.29% of cases.

2. TASK AND DATA

For two clips (a_A, a_B) rendering transcript x , the judge produces a rationale c_d and scores $(s_d^A, s_d^B) \in \{1, \dots, 10\}^2$ for dimension d . Its per-dimension verdict is

$$v_d = \text{sgn}(s_d^A - s_d^B) \in \{A, \text{Tie}, B\}. \quad (1)$$

Table 1 summarizes the seven dimensions. LPJ always evaluates C1–C4, activates L1–L3 from transcript cues, and emits an overall verdict; D-LPJ emits only C1–C4, with aggregation performed externally. All experiments use the high-expressiveness profile.

The 1,043-pair main suite comprises Qwen3-TTS [14] base versus our SFT variant (T1, 417; T1* is its 119-pair unanimous subset), two samples from our internal Qwen3-TTS-GDPO variant (T2, 111 unanimous) testing within-model variation, the same internal variant versus BERT-CosyVoice [15] (T3, 315), and human recordings versus Qwen3-TTS (T4, 200). T5 adds 222 IndexTTS2 [16]-versus-FireRedTTS [17] transfer pairs from a model combination absent from coupled-judge training.

T1–T4 include 135 speakers across warm-recommendation, passionate-promotion, and professional-explanation styles, with matched product categories. Transcripts combine real-stream ASR and generated copy for product entry, explanation, and ordering; durations are balanced across 5–10, 10–20, and 20–30 seconds. Each pair shares one transcript, reducing lexical confounding across cross-model, same-model, and human-versus-TTS comparisons.

Three trained contractors independently label each pair using the teacher’s rubric definitions and pairwise criteria, blinded to system and source. Across T1–T4, 45% of labels are unanimous 3–0 and 55% have two agreeing votes and one abstention; exclusions are limited to corrupted audio, transcript mismatch, or protocol violations. Frozen evaluation partitions do not overlap training or validation in audio, pair, normalized transcript, speaker, or source stream. C1–C4 test sets contain 224, 172, 142, and 160 unanimous non-tie pairs. Audio is not redistributed because of licensing.

3. METHODOLOGY

3.1. Swap-consistent distillation and curriculum

Gemini judges each training pair twice in both A/B orders; swapped votes are mapped back to the clips. LPJ retains only 4–0 outcomes and balances the winner between slots. Its curriculum moves from human-versus-TTS to cross- and same-model comparisons. We attach rank-32 LoRA [18] to Qwen3-Omni attention projections and merge it after SFT.

For section token set T_d and dimension weight α_d , the weighted SFT loss is

$$\mathcal{L}_{\text{SFT}} = - \frac{\sum_d \alpha_d \sum_{t \in T_d} \log \pi_\theta(y_t | y_{<t}, a_A, a_B, x)}{\sum_d \alpha_d |T_d|}. \quad (2)$$

The coupled judge uses weights (1.5, 1.5, 1.5, 1.5, 1.5, 0.7, 0.7), ordered as (C1, C2, C3, C4, L1, L2, L3), with weights 1.0 and 2.0 on the summary and conclusion, respectively.

3.2. Dimension-wise supervision

D-LPJ queries Gemini four times per core dimension, with each call returning only that dimension’s rationale and score pair. After restoring swapped votes, we retain 4–0, 3–1, or three agreeing non-tie

votes plus one abstention. Other pair-dimensions receive zero SFT loss while confident dimensions remain trainable. Retention therefore occurs at the pair-dimension level instead of dropping an entire pair when one label is uncertain. Retained sections form one C1–C4 rationale target without a summary or overall conclusion.

Both D-LPJ SFT stages use rank-64 QLoRA and learning rate 5×10^{-5} , for 8 and 14 epochs, respectively, with batch size 1 and eight-step gradient accumulation.

For GRPO [13], $G = 8$ completions are sampled for prompt q . For any group-level signal z_i , we use

$$\mathcal{N}_G(z_i) = \frac{z_i - \text{mean}_j(z_j)}{\text{std}_j(z_j) + \epsilon}. \quad (3)$$

The coupled Live-ProsodyJudge receives one reward: +1 when its overall verdict matches the target and -1 otherwise. Its normalized advantage is broadcast over the full completion. For D-LPJ, external aggregation is $V_w = \text{sgn}[\sum_d w_d(s_d^A - s_d^B)]$, with zero treated as a tie. The aggregate-reward baseline uses $w = (1, 1, 1, 1)$ and broadcasts the resulting single verdict reward over the completion, even though the model emits no conclusion.

Span-local GRPO instead defines, for every confident dimension,

$$r_{i,d} = \begin{cases} +1, & v_{i,d} = \tilde{v}_d, \\ -1, & \text{otherwise,} \end{cases} \quad d \in \{C1, \dots, C4\}, \quad (4)$$

and normalizes each dimension independently across the same rollout group. Its token-level advantage is

$$\hat{A}_{i,t} = \begin{cases} \mathcal{N}_G(r_{i,d}), & t \in T_{i,d} \text{ and } d \text{ is confident,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Consequently, C1 reward cannot directly cancel C4 reward in the token objective, although all dimensions share model parameters. If every rollout receives the same reward for one dimension, normalization gives zero advantage and that dimension contributes no policy gradient for the group. Both GRPO variants start from the same two-stage D-LPJ SFT checkpoint and use a new rank-64 LoRA, learning rate 10^{-5} , reverse-KL coefficient 0.04, and three GRPO epochs. We add no format reward.

3.3. Decoupling design and evaluation

D-LPJ reduces cross-dimension coupling in both targets and optimization. Independent teacher queries generate dimension-specific rationales without an overall verdict; pair-dimension masking retains reliable supervision even when other dimensions of the same pair are uncertain. Removing the shared conclusion keeps SFT focused on the four dimension outputs, while span-local GRPO assigns each dimension’s reward only to its own rationale. Together, these mechanisms encourage distinct judgments and dimension-specific credit assignment.

Here, independent judgments refer to separate dimension-level decisions rather than complete statistical independence: the four dimensions still share the same audio, transcript, autoregressive context, and model parameters, and genuine prosodic correlations may remain. We evaluate the resulting judgments in two complementary settings. In the 140-pair multidimensional human set, three annotators assess C1–C4 for every audio pair, with confidence determined separately for each pair-dimension. The larger dimension-specific sets provide additional high-confidence human labels for measuring accuracy on each individual axis.

Table 1. Seven-dimension live streaming prosody rubric. “Text” means that the dimension is evaluated only when the transcript contains the corresponding cue.

ID	Dimension	Gate	Principal attributes
C1	Fluency and Naturalness	Always	Rate stability, phrasing, pauses, coherence
C2	Intonation Variation	Always	Pitch range, final contours, stress, sentence contrast
C3	Emotional Expression	Always	Emotion-content match, intensity, warmth and appeal
C4	Live Streaming Expressiveness	Always	Energy, rate drive, rhythm, real-host quality
L1	Key Information Emphasis	Text	Prominence of price, discount, scarcity, and calls to action
L2	Emotional State Switching	Text	Capture and smoothness of emotional turning points
L3	Interaction Mode Switching	Text	Delivery change between explanation and audience address

4. EXPERIMENTS

4.1. Setup and coupled-judge results

Models are trained with SWIFT [19] in bf16 on four 72GB GPUs. All reported training comparisons use one fixed random seed. Gemini uses one judgment in the presented order, reflecting its intended use and the cost of repeated proprietary inference. SpeechJudge-GRM and our students use ten sampled judgments, five in each A/B order. Swapped score pairs are restored to physical clips, each clip’s scores are averaged across the ten generations, and the verdict is the sign of the mean score difference. A tie is counted as an error.

We report agreement with the final human A/B label. For paired system comparisons, we draw 10,000 bootstrap resamples of test-pair indices and apply identical indices to both systems; the 2.5th and 97.5th percentiles of the accuracy-difference distribution define the 95% interval. These intervals quantify test-pair sampling uncertainty for the fixed checkpoints, not variation across training seeds.

Table 2 shows that LPJ v1+GRPO has higher point accuracy than single-call Gemini in every column. Relative to v1 SFT, its changes are +0.48, +1.68, 0.00, +0.63, and +3.50 points on T1, T1*, T2, T3, and T4. Table 3 reports the paired intervals. Every interval includes zero, so these are descriptive changes. Admitting lower-confidence 3-1 labels (v0) is worse than 4-0-only v1 on four of five columns, indicating that confidence-filtered supervision is more important than simply retaining more teacher-labeled data. Removing the large-gap first stage is associated with a 34.5-point lower T4 estimate, while single-stage training is lower on T2-T4 but higher on T1.

On T5, SpeechJudge-GRM obtains 185/222 (83.33%), Gemini obtains 189/222 (85.14%), and LPJ v1+GRPO obtains 192/222 (86.49%). The three-pair, 1.35-point gap over Gemini has no paired confidence interval and is treated only as descriptive transfer evidence.

4.2. Position-bias diagnostic

We measure position bias using all 4,170 judgments from the 417 T1 pairs under balanced presentation. Each physical comparison occurs in both A/B assignments, so pooling assignments separates presentation slot from clip identity. Let $\Delta_{\text{pos}} = \bar{s}_B - \bar{s}_A$ denote the difference between scores assigned to the second- and first-presented slots. We also report the fraction of raw judgments whose verdict favors presented B; the t statistic is computed from pair-level mean slot differences.

Table 4 shows that SpeechJudge-GRM assigns the second-presented clip nearly one extra point and favors B in two thirds of judgments. In contrast, Live-ProsodyJudge is close to slot symmetry in both score gap and win rate. This is consistent with swap-

consistency filtering and winner-balanced training order, but it is not a causal ablation of either component. Raw score levels are also not compared across models; the diagnostic concerns within-model A/B asymmetry.

4.3. Dimension-wise evaluation

Across the evaluation sets, D-LPJ produces at least two different non-tie core-dimension verdicts on 24.3–60.6% of pairs, showing that it can avoid unanimous verdict vectors. We additionally use a 140-pair multidimensional human test set. For each audio pair, three trained annotators independently label C1–C4; confidence is determined separately for every pair-dimension, so not every pair has confident labels on all four axes. On this set, the coupled LPJ produces 0/140 non-unanimous core vectors and 48.0% agreement on the confident human dimension labels. D-LPJ SFT produces 90/140 non-unanimous vectors and 67.9% agreement; span-local GRPO changes these to 100/140 (71.4%) and 71.4%, respectively. This same-set comparison evaluates both resistance to verdict collapse and agreement with the retained human judgments.

Table 5 therefore evaluates each dimension against a separate, unanimous human test set. Ten-sample SFT exceeds one-call Gemini in every dimension. Span-local GRPO changes pooled agreement from 73.64% to 77.94% with one sample and from 84.10% to 86.10% with ten samples. At ten samples it is 1.00 point above aggregate-reward GRPO. Table 6 shows that all per-dimension paired intervals for span-local GRPO versus SFT and aggregate-reward GRPO include zero; we therefore report point estimates without claiming statistical significance.

4.4. Additional diagnostics

Repeated, balanced-order inference contributes unevenly. Single-sample LPJ v1+GRPO obtains 67.87, 83.19, 65.77, 80.95, and 74.50% on T1, T1*, T2, T3, and T4; ten-sample aggregation changes these estimates by +3.35, −0.84, +4.50, +1.91, and +9.00 points. It combines repeated generation with order balancing, whose effects are not isolated here.

D-LPJ also permits external scalar decisions without restoring an overall text target. Validation-selected weights $w = (3, 0.5, 0.5, 3)$ give 73.6% on T1 and 85.5% on T4, versus 69.3% and 78.5% with uniform weights; T2 remains 71.2%. This is not a causal comparison with the coupled judge because their objectives differ.

We also test CEAEval’s adaptive audio-attention bias [7]. A representative instance contains 426 audio, 634 rationale, and 2,618 rubric-prompt tokens. It trains stably but does not improve held-out accuracy, so it is excluded from the final model. One possible explanation is that the generated rationale is not much longer than the

Table 2. Overall human-label agreement (%). “1s” denotes the first sampled judgment and “10s” denotes 10-sample balanced-order aggregation; Gemini uses one call. T1★ is part of T1.

Model	T1	T1★	T2	T3	T4
SpeechJudge-GRM (10s)	55.88	72.27	48.65	53.97	42.00
Gemini-3.1-pro-preview (1s)	65.23	72.27	65.77	73.33	58.00
Without stage-1 SFT (10s)	53.00	72.27	60.36	69.84	45.50
Single-stage SFT (10s)	72.66	80.67	67.57	80.00	74.50
LPJ v0, 4-0 + 3-1 (10s)	71.22	78.99	65.77	80.63	73.00
LPJ v1, 4-0 SFT (1s)	64.99	74.79	63.06	73.97	66.50
LPJ v1, 4-0 SFT (10s)	70.74	80.67	70.27	82.22	80.00
LPJ v1 + GRPO (1s)	67.87	83.19	65.77	80.95	74.50
LPJ v1 + GRPO (10s)	71.22	82.35	70.27	82.86	83.50

Table 3. Paired-bootstrap comparison of 10-sample LPJ v1+GRPO against v1 SFT. Differences and 95% intervals are in percentage points.

Set	GRPO – SFT	95% interval
T1	+0.48	[−2.64, +3.60]
T1★	+1.68	[−2.52, +6.72]
T2	0.00	[−7.21, +7.21]
T3	+0.63	[−1.90, +3.18]
T4	+3.50	[−1.50, +8.50]

Table 4. Position statistics on T1 under 10-sample balanced-order inference.

Model	$\bar{s}_A$	$\bar{s}_B$	Δ_{pos}	t	B win
Live-ProsodyJudge	7.842	7.854	+0.012	+0.4	50.7%
SpeechJudge-GRM	7.061	8.039	+0.978	+11.7	66.8%

audio sequence, limiting the benefit of explicitly up-weighting audio attention; substantially longer generations may behave differently.

4.5. Best-of-8 and limitations

For each transcript, the TTS model generates eight candidates and Live-ProsodyJudge evaluates all $\binom{8}{2} = 28$ distinct pairs. Without requiring absolute-score calibration, we select the candidate with the largest number of pairwise wins,

$$\hat{a} = \arg \max_{a_i} \sum_{j \neq i} \mathbf{1}[V(a_i, a_j) = a_i]. \quad (6)$$

Two trained contractor annotators independently rank the top three candidates for 400 sets. We retain the 136 sets for which both produce the same ordered top-three ranking; this inclusion decision is made without model predictions. On this high-confidence subset, the selector obtains Hit@1/2/3 of 98/136 (72.06%), 106/136 (77.94%), and 116/136 (85.29%), compared with random-selection baselines of 12.5%, 25.0%, and 37.5%. Hit@ k means that the tournament winner lies within the agreed human top- k . These rates characterize the retained subset and do not estimate performance on all 400 collected sets.

Three limitations bound the evidence. First, training results use one seed; bootstrap intervals quantify test-pair sampling uncertainty, not variation across training runs. Second, ten-sample comparisons

Table 5. D-LPJ agreement with unanimous human dimension labels (%). “Agg.” and “Span” denote aggregate-reward and span-local GRPO.

Model	C1	C2	C3	C4	All
Gemini (1s)	74.55	77.33	80.28	84.38	78.65
SFT (1s)	68.75	65.70	76.06	86.88	73.64
SFT (10s)	79.02	80.81	85.92	93.13	84.10
Agg. GRPO (1s)	70.54	71.51	79.58	90.62	77.22
Agg. GRPO (10s)	79.91	83.14	87.32	92.50	85.10
Span GRPO (1s)	71.88	76.74	76.76	88.75	77.94
Span GRPO (10s)	83.04	83.14	85.21	94.38	86.10

Table 6. Ten-sample span-local GRPO differences and paired-bootstrap 95% intervals, in percentage points.

Dim.	Versus SFT	Versus Agg. GRPO
C1	+4.02 [−0.89, +9.38]	+3.12 [−1.79, +7.59]
C2	+2.33 [−2.33, +6.98]	0.00 [−5.23, +5.23]
C3	−0.70 [−4.23, +2.82]	−2.11 [−6.34, +2.11]
C4	+1.25 [−1.88, +4.38]	+1.88 [−1.88, +5.62]

against Gemini use more inference and balanced order. Third, although the 140-pair multidimensional set provides complete human C1–C4 annotations, most pairs in the larger dimension-specific test sets do not include complete annotations across all four dimensions. D-LPJ is currently limited to C1–C4, and using its vector rewards to optimize a TTS model remains future work.

5. CONCLUSION

We presented LPJ and D-LPJ for multi-dimensional live streaming prosody evaluation. Balanced-order LPJ achieves strong point agreement, reduces measured slot bias relative to SpeechJudge-GRM, and supports Best-of-8 selection. D-LPJ removes the shared overall target and localizes SFT and GRPO supervision by dimension, producing varied core-dimension judgments with 86.10% pooled agreement. Together, the two judges provide scalable candidate ranking and dimension-specific diagnosis. Future work will extend decoupling to sparse conditional dimensions and use dimension-wise rewards for TTS post-training.

6. FUNDING ACKNOWLEDGMENT

This work was supported by the TaoLive-AIGC Team, Taobao & Tmall Group of Alibaba.

7. COMPLIANCE WITH ETHICAL STANDARDS

Human labels were produced by contracted annotation staff who received task-specific training and were blinded to system identity. We report only aggregate results. Evaluation audio is not publicly redistributed because source recordings are subject to copyright and licensing restrictions.

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